MINI PROJECT REPORT

ON

RESPIRATORY DISEASE PREDICTION USING MACHINE LEARNING

Lung Auscultation Sound Classification

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING (DS)

Submitted by

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To be an exceptional knowledge-driven institution advancing a culture of honesty and compassion to make a difference to the world.

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PROBLEM STATEMENT

Respiratory sounds recorded through lung auscultation may contain acoustic patterns associated with different respiratory conditions. Manually interpreting these recordings can be challenging, especially when the dataset contains many recordings and multiple disease categories. This project develops a machine-learning pipeline that processes respiratory sound recordings, extracts MFCC-based features and compares four classification models—Support Vector Machine (SVM), Random Forest, Artificial Neural Network (ANN) and Convolutional Neural Network (CNN). The system is intended for academic experimentation and is not a substitute for professional medical diagnosis.

Project Objectives

- Load and inspect respiratory sound recordings and their diagnosis labels.
- Apply audio noise reduction, MFCC feature extraction and normalization.
- Train and evaluate SVM, Random Forest, ANN and CNN models.
- Compare model performance using accuracy, precision, recall and F1-score.
- Visualize results using confusion matrices, ROC curves and precision–recall curves.

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CHAPTER 1: INTRODUCTION

1.1 Introduction to the Project

Respiratory Disease Prediction using Machine Learning is an audio-classification project based on lung auscultation recordings. The workflow converts each recording into numerical features and uses these features to train models that assign a respiratory-disease label. The project compares traditional machine-learning methods with neural-network approaches.

1.2 Background

A respiratory recording is a time-varying signal. Its frequency content and acoustic texture may differ across recordings. Machine-learning models cannot directly use raw audio in the same way a person listens to it, so audio preprocessing and feature extraction are used to represent the signal numerically. Mel-Frequency Cepstral Coefficients (MFCCs) summarize aspects of the sound's spectral envelope and are widely used in audio analysis.

1.3 Motivation

The project provides practical experience in applying signal processing, feature engineering, classification and evaluation to a real-world dataset. It also demonstrates why model evaluation must consider class distribution instead of relying on accuracy alone.

1.4 Need for the Project

A reproducible pipeline makes it possible to compare several algorithms using the same dataset split and evaluation measures. Such comparison is useful for learning and experimentation, while the limitations of the dataset and models must be stated clearly.

1.5 Objectives

- Perform noise reduction on respiratory audio.
- Extract and normalize MFCC features.
- Train SVM, Random Forest, ANN and CNN models.
- Compare the models on a held-out test set.
- Generate visual evaluation outputs for interpretation.

1.6 Organization of the Report

Chapter 1 introduces the project and its objectives. Chapter 2 summarizes the technical background and related approaches. Chapter 3 explains the dataset, preprocessing and models. Chapter 4 describes the system workflow and implementation requirements.

CHAPTER 2: LITERATURE SURVEY

2.1 Literature Survey

Respiratory sound classification has been studied using signal-processing features and machine-learning classifiers. The ICBHI 2017 respiratory sound database provides recordings and annotations intended to support research in respiratory sound analysis. Common audio-analysis workflows extract features such as MFCCs and evaluate classifiers using held-out data.

Traditional classifiers such as Support Vector Machines and Random Forests can operate on compact numerical feature vectors. Artificial Neural Networks learn nonlinear relationships through interconnected layers, while Convolutional Neural Networks can learn local patterns from structured time-frequency representations.

A fair comparison requires a consistent train-test split, clearly defined evaluation metrics and attention to class imbalance. A high overall accuracy can coexist with weak performance on rare classes, so confusion matrices and class-wise precision and recall are important supporting evidence.

2.2 Summary

Based on this background, the project implements SVM, Random Forest, ANN and CNN using MFCC-based representations and compares their results using the same held-out test set. This report describes the implemented work and does not claim that the models are clinically validated.

CHAPTER 3: METHODOLOGY

3.1 Introduction

The methodology consists of dataset inspection, audio preprocessing, feature extraction, data splitting, model training and evaluation. Two MFCC representations are used: a fixed-length feature vector for SVM, Random Forest and ANN, and a time-frequency MFCC matrix for CNN.

3.2 Existing System

Respiratory sound interpretation may be performed by listening to recordings and reviewing clinical information. Manual interpretation requires expertise and can be difficult to scale across large collections. Automated classification can support analysis and research, but its output must not be treated as a standalone medical decision.

3.3 Proposed Work

The proposed academic system uses audio preprocessing and four classification models to compare respiratory disease-label prediction. It begins with the provided Respiratory Sound Database and ends with evaluation tables and plots.

3.3.1 Data Collection

The project uses the Respiratory Sound Database, containing 920 WAV recordings and 920 associated annotation text files, with diagnosis information for 126 patients. The audio and annotation files are kept in the supplied nested directory structure. The diagnosis file is used to map recording patient identifiers to disease labels.

Dataset item	Details
Audio recordings	920 WAV files
Annotation files	920 TXT files
Patients represented	126
Original patient-level diagnosis categories	COPD, Healthy, URTI, Bronchiectasis, Pneumonia, Bronchiolitis, LRTI and Asthma
Usable samples for stratified model split	919; one Asthma sample was excluded because that class contained only one sample

3.3.2 Data Pre-processing and Feature Extraction

Audio loading

Audio files are loaded using librosa while retaining the original sampling rate. A sample recording was observed at 44,100 Hz with 882,000 samples, corresponding to approximately 20 seconds.

Noise reduction

The noisereduce library is applied to reduce unwanted background noise. Noise reduction modifies the signal values but does not necessarily change the number of samples or the recording duration.

MFCC extraction

Thirteen MFCC coefficients are extracted from each audio signal. MFCCs provide a compact numerical representation of spectral characteristics over time.

Normalization and feature preparation

The extracted MFCC values are normalized using the mean and standard deviation. For SVM, Random Forest and ANN, the normalized MFCC matrix is averaged across time to create a 13-value feature vector. For CNN, the MFCC matrix is retained as a two-dimensional time-frequency representation and padded or truncated to a consistent number of frames.

Train-test split

The usable dataset is divided into training and testing subsets using an 80:20 split, a fixed random state and stratification. The resulting shapes are 735 training samples and 184 testing samples for the fixed-vector models.

3.4 Algorithms Used

3.4.1 Support Vector Machine (SVM)

SVM identifies a decision boundary that separates classes in the feature space. It is used here as a traditional machine-learning baseline on the 13-value MFCC feature vectors.

3.4.2 Random Forest

Random Forest combines the predictions of multiple decision trees. The project uses 100 trees and a fixed random state to support reproducibility.

3.4.3 Artificial Neural Network (ANN)

The ANN is a feed-forward network with dense layers. The implemented architecture uses an input of 13 features, hidden layers with 64 and 32 units using ReLU activation, and a softmax output layer for the seven retained classes.

3.4.4 Convolutional Neural Network (CNN)

The CNN receives the MFCC matrix as a single-channel input. Convolution and max-pooling layers learn local patterns, followed by a dense layer and a softmax output layer. The model is trained to classify the retained disease categories.

CHAPTER 4: SYSTEM DESIGN AND IMPLEMENTATION

4.1 System Design

The system is organized as a sequential pipeline: data collection, preprocessing, feature extraction, dataset splitting, model training, prediction and evaluation.

Module	Function
Data collection	Reads WAV files, annotations and patient diagnosis labels.
Preprocessing	Loads audio and applies noise reduction.
Feature extraction	Calculates and normalizes MFCC features.
Model training	Fits SVM, Random Forest, ANN and CNN models.
Evaluation	Calculates metrics and creates confusion matrices, ROC curves and precision–recall curves.
Result storage	Saves trained models, the comparison table and evaluation figures.

4.2 Requirements

4.2.1 Software Requirements

- Python 3.13.1
- Visual Studio Code and Jupyter Notebook
- NumPy and pandas for numerical and tabular data handling
- librosa and noisereduce for audio processing
- scikit-learn for data splitting, traditional models and evaluation
- TensorFlow/Keras for ANN and CNN
- Matplotlib and Seaborn for visualizations

4.2.2 Hardware Requirements

A computer capable of running Python and the listed libraries is required. Sufficient RAM and storage are needed for the audio dataset, feature arrays, model files and generated plots. A dedicated GPU is not required by the implemented project, although training time may vary with hardware.

4.3 Data Flow

1. Read the audio recordings and diagnosis labels.

2. Apply noise reduction to the audio signal.
3. Extract 13 MFCC coefficients and normalize the values.
4. Prepare fixed-length feature vectors for SVM, Random Forest and ANN; prepare MFCC matrices for CNN.
5. Split the usable samples into training and testing sets.
6. Train each model using the training set and predict labels for the test set.
7. Calculate evaluation metrics and save comparison results and plots.

4.4 Project Implementation Details

The implementation was developed in a Jupyter Notebook within Visual Studio Code. The project directory separates the dataset, notebook, saved models and results. The dataset folder is excluded from version control because of its size.

4.5 Reporting and Visualization

- Confusion matrices summarize correct and incorrect predictions by class.
- ROC curves compare true-positive and false-positive rates across thresholds.
- Precision–recall curves show the trade-off between precision and recall.
- A CSV comparison table stores accuracy, weighted precision, weighted recall and weighted F1-score.

CHAPTER 5: RESULTS AND DISCUSSION

5.1 Model Performance

The final test-set accuracy values obtained after the required preprocessing workflow are summarized below.

Model	Test accuracy
Support Vector Machine (SVM)	86.41%
Random Forest	86.96%
Artificial Neural Network (ANN)	86.41%
Convolutional Neural Network (CNN)	86.96%

Random Forest and CNN achieved the highest recorded overall accuracy of 86.96%, followed by SVM and ANN at 86.41%. The difference between these accuracy values is small.

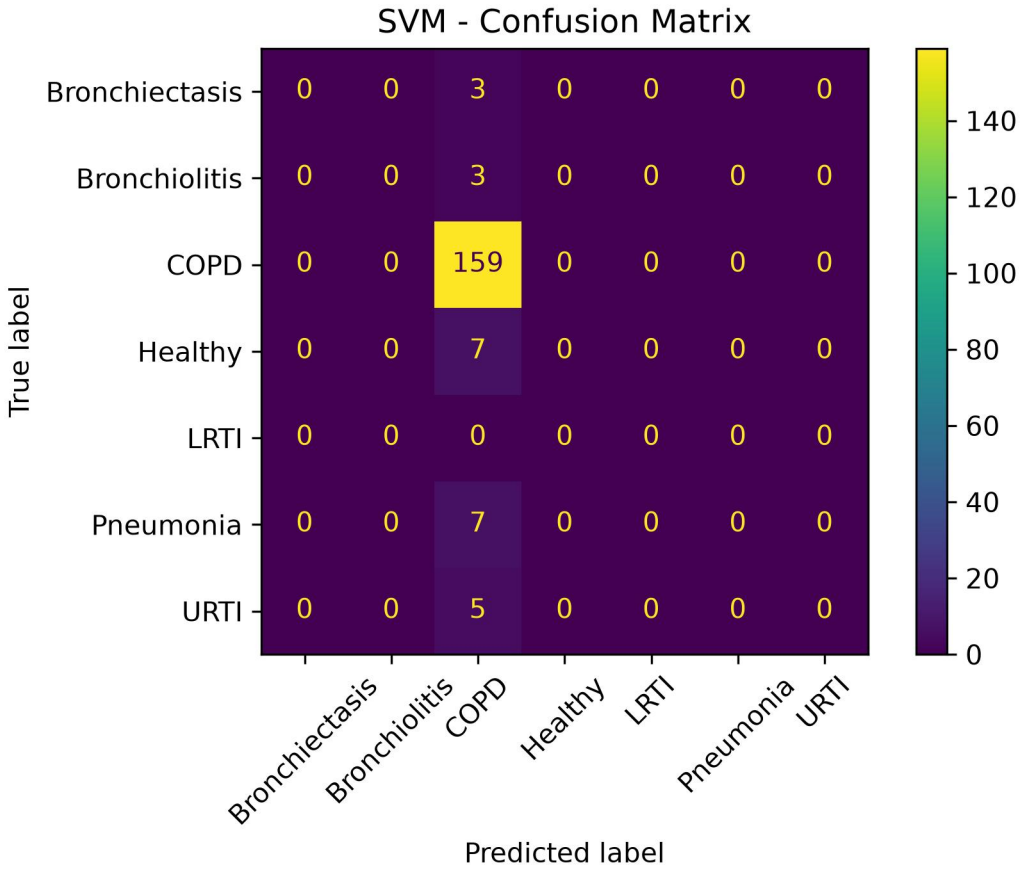
5.2 Results Interpretation

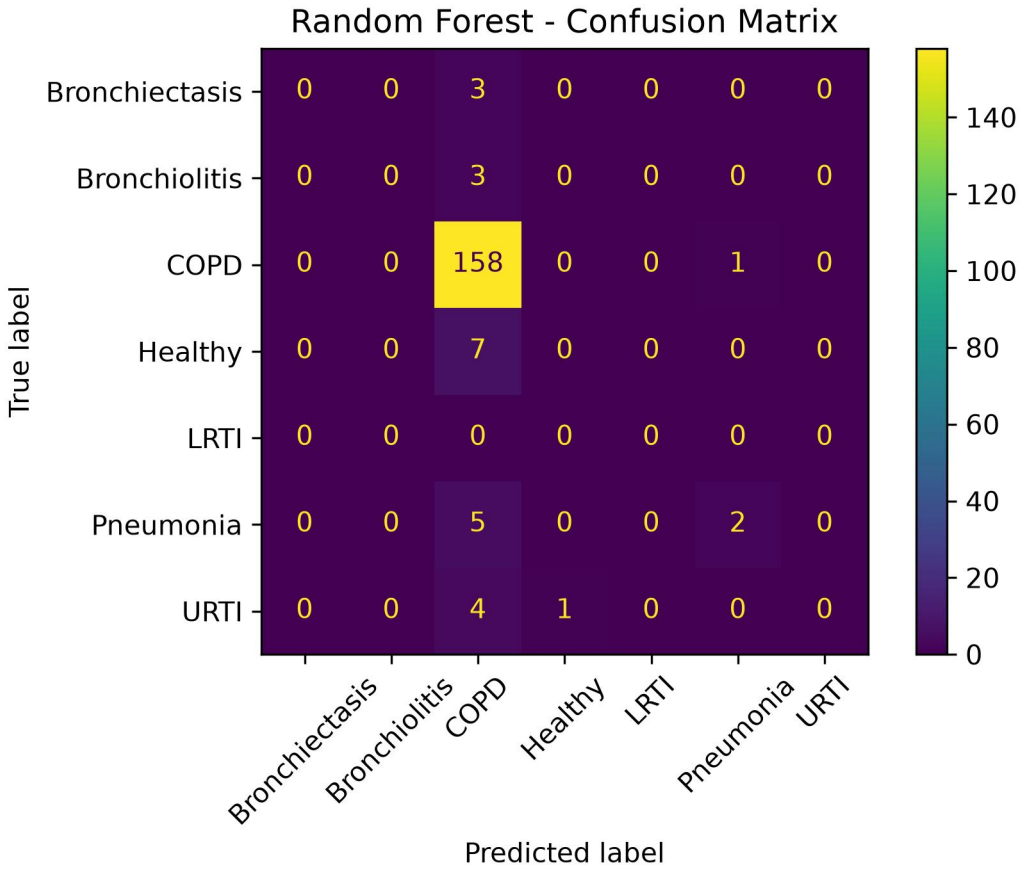
The dataset is highly imbalanced: COPD contributes the majority of audio files, while several other disease categories contain very few examples. Consequently, a model can achieve high overall accuracy by correctly predicting the majority class while performing poorly on rare classes. The final SVM report showed that its predictions were concentrated on COPD. The Random Forest report also showed substantially stronger performance on COPD than on several minority categories.

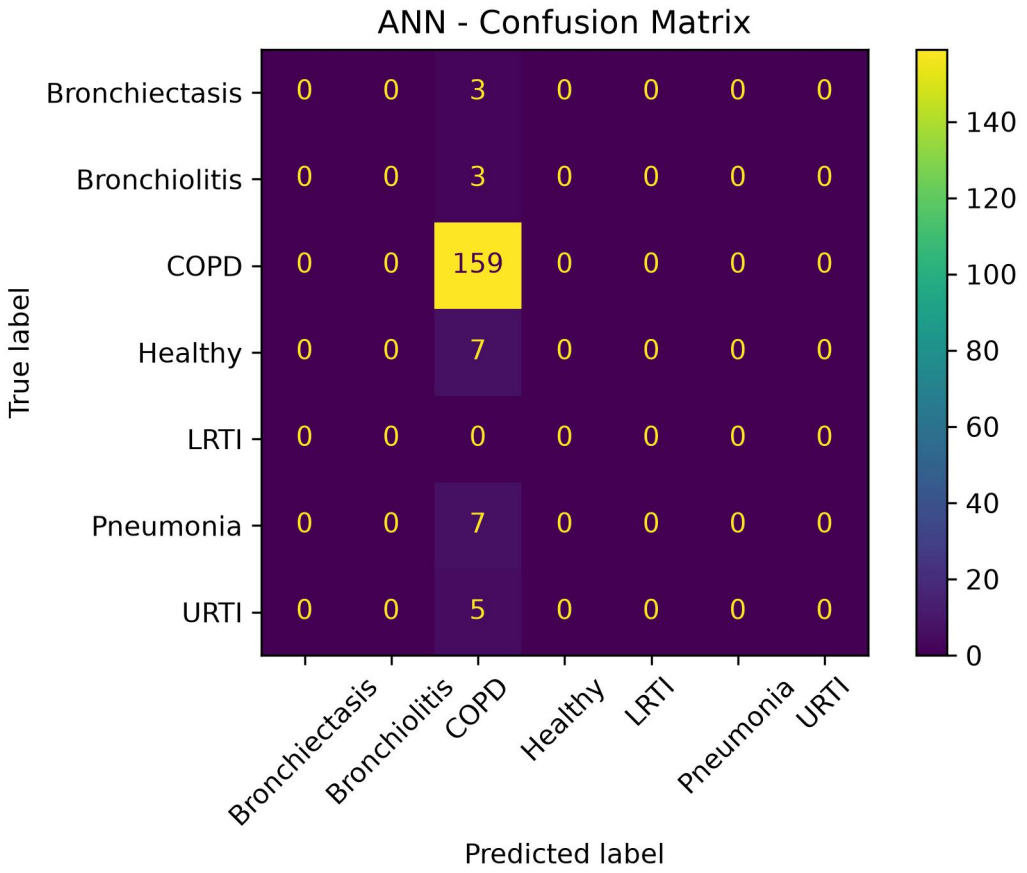
For this reason, the accuracy values should be considered alongside weighted precision, weighted recall, weighted F1-score, class-wise metrics and confusion matrices. The results demonstrate the behavior of these models on this dataset split; they do not establish clinical reliability.

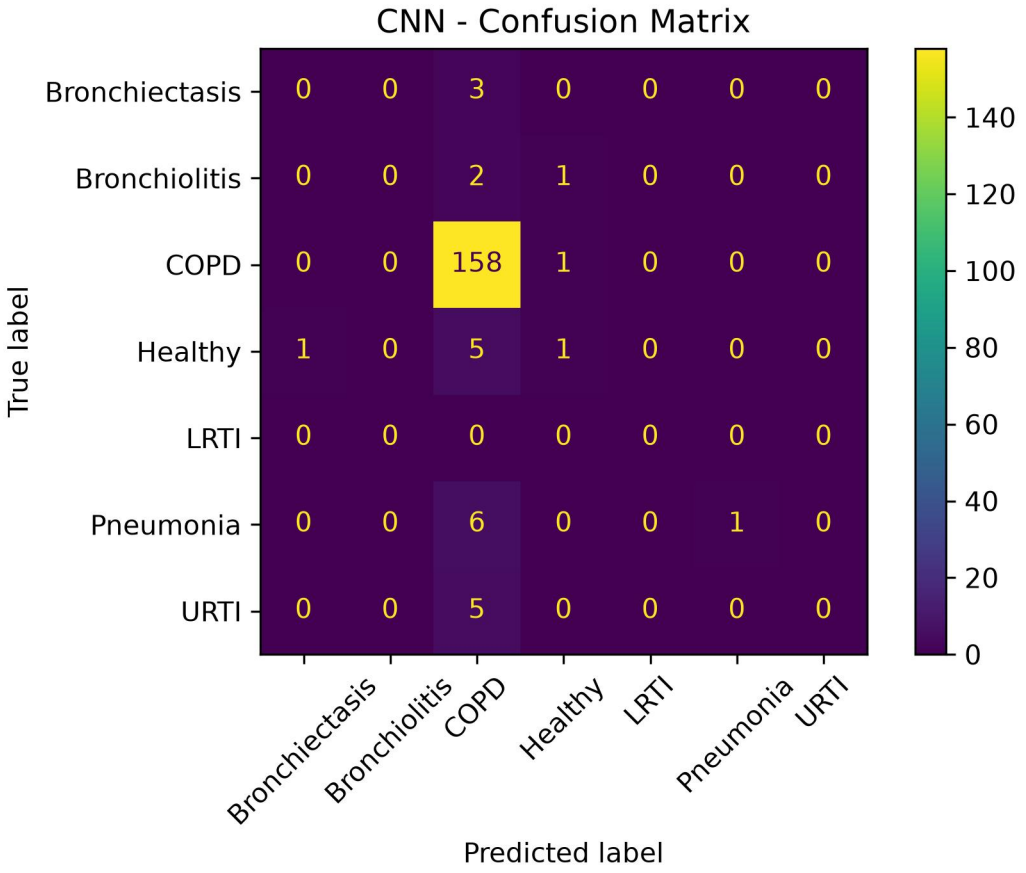
5.3 Confusion Matrices

Insert the final confusion-matrix figures generated by the notebook. Use one clearly captioned figure for each model and ensure that the class labels remain readable.


SVM - Confusion Matrix
Bronchiectasis
Bronchiolitis
COPD
Healthy
LRTI
Pneumonia
URTI
True label
Predicted label
0 0 3 0 0 0 0
0 0 3 0 0 0 0
0 0 159 0 0 0 0
0 0 7 0 0 0 0
0 0 0 0 0 0 0
0 0 7 0 0 0 0
0 0 5 0 0 0 0
140
120
100
80
60
40
20
0

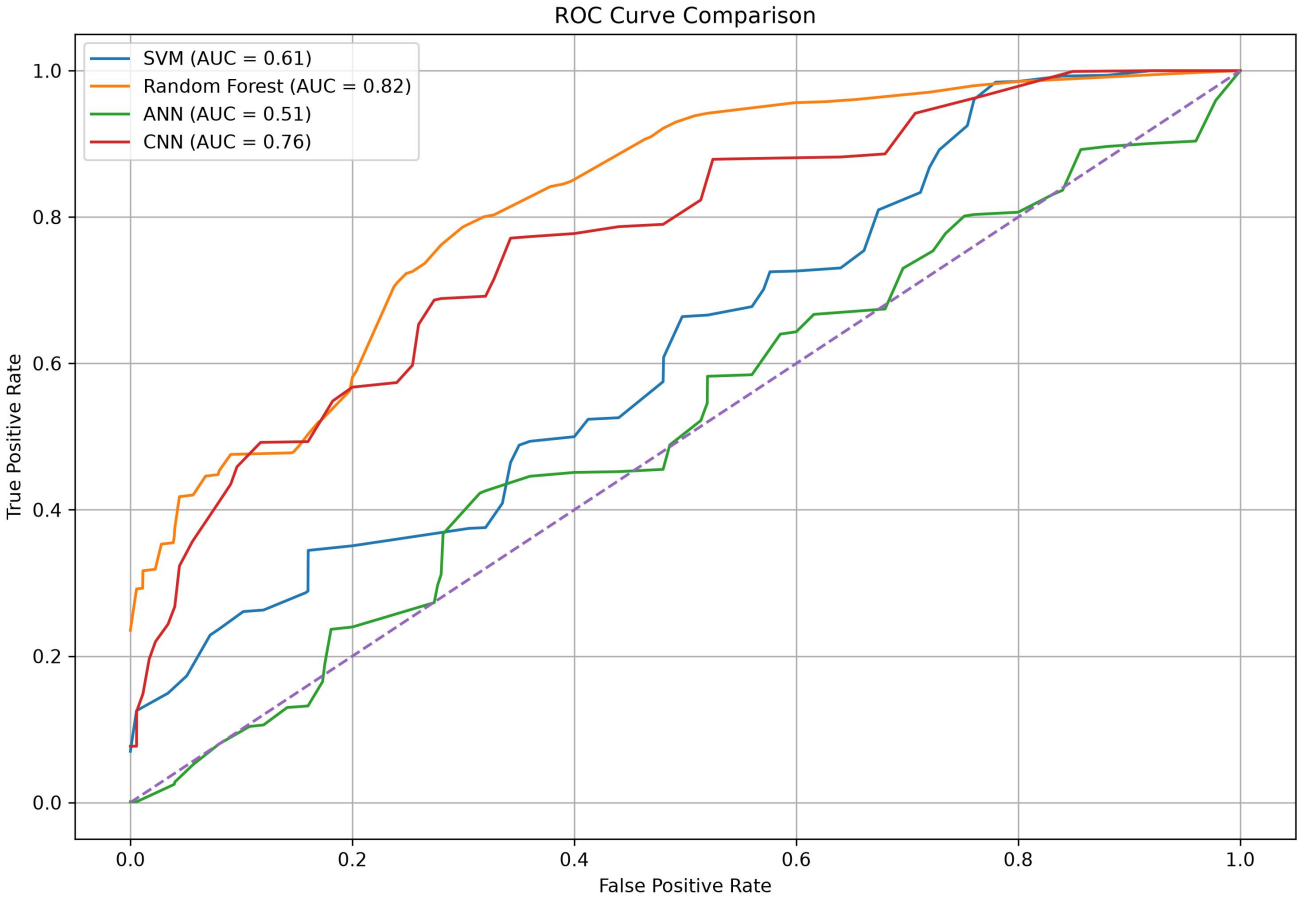

Random Forest - Confusion Matrix
Bronchiectasis
Bronchiolitis
COPD
Healthy
LRTI
Pneumonia
URTI
True label
Predicted label
0 0 3 0 0 0 0
0 0 3 0 0 0 0
0 0 158 0 0 1 0
0 0 7 0 0 0 0
0 0 0 0 0 0 0
0 0 5 0 0 2 0
0 0 4 1 0 0 0
140
120
100
80
60
40
20
0


ANN - Confusion Matrix
Bronchiectasis
Bronchiolitis
COPD
Healthy
LRTI
Pneumonia
URTI
True label
Predicted label
0 0 3 0 0 0 0
0 0 3 0 0 0 0
0 0 159 0 0 0 0
0 0 7 0 0 0 0
0 0 0 0 0 0 0
0 0 7 0 0 0 0
0 0 5 0 0 0 0
140
120
100
80
60
40
20
0



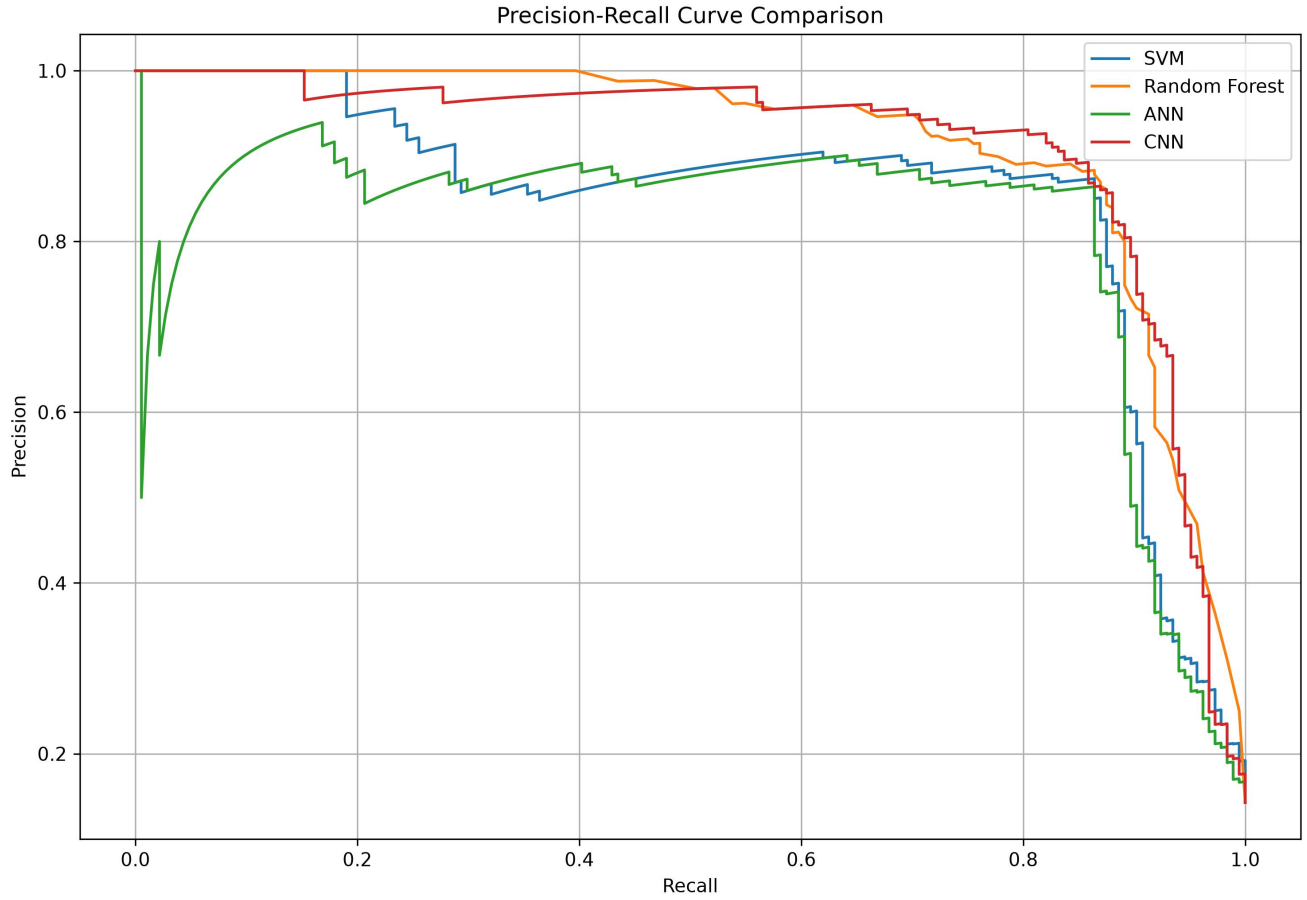
5.4 ROC Curve

Insert the final ROC curve comparison generated for the four models. The ROC curve visualizes the relationship between true-positive rate and false-positive rate at different thresholds. Classes without positive examples in the test set cannot provide a class-specific ROC curve.



5.5 Precision–Recall Curve

Insert the final precision–recall curve comparison generated by the notebook. This plot is particularly useful when class frequencies are uneven, although the interpretation should consider the method used to aggregate the classes.



5.6 Summary

The final models were trained and evaluated, and the required evaluation plots and comparison CSV were saved. Random Forest and CNN shared the highest overall accuracy in the recorded results. The main limitation is the imbalance in the dataset and the small number of recordings for several disease classes.

CHAPTER 6: CONCLUSION AND FUTURE WORK

6.1 Conclusion

This mini project implemented a machine-learning pipeline for classifying respiratory sound recordings using the Respiratory Sound Database. The workflow included audio loading, noise reduction, MFCC extraction, normalization, feature preparation, model training and evaluation. SVM, Random Forest, ANN and CNN were compared using a held-out test set.

Random Forest and CNN achieved the highest final accuracy of 86.96%, while SVM and ANN achieved 86.41%. However, the dataset was strongly imbalanced, and the results indicate that performance on the majority COPD class was better than performance on several minority classes. Therefore, the project is an academic classification experiment rather than a clinically validated diagnostic tool.

6.2 Future Work

- Evaluate the pipeline on a larger and more balanced dataset.
- Improve the representation and evaluation of minority disease classes.
- Use patient-level separation between training and testing to better assess generalization to unseen patients.
- Review the preprocessing and validation pipeline and repeat experiments with consistent settings across all models.
- Consider a user interface only if it becomes a required extension of the project.

REFERENCES

8. Rocha, B. M., et al. (2019). A respiratory sound database for the development of automated classification. In Proceedings of the International Conference on Biomedical and Health Informatics (ICBHI).
9. Librosa Development Team. librosa: Audio and music signal analysis in Python. https://librosa.org/
10. scikit-learn developers. scikit-learn: Machine learning in Python. https://scikit-learn.org/
11. TensorFlow team. TensorFlow documentation. https://www.tensorflow.org/
12. noisereduce contributors. noisereduce: Noise reduction in audio signals. https://github.com/timsainb/noisereduce

APPENDIX A: PROJECT OUTPUTS

The project directory contains the notebook, trained model files, evaluation figures and model comparison CSV. The following outputs were generated:

- svm_model.pkl and random_forest_model.pkl
- ann_model.keras and cnn_model.keras
- label_encoder.pkl
- svm_confusion_matrix.png
- random_forest_confusion_matrix.png
- ann_confusion_matrix.png
- cnn_confusion_matrix.png
- roc_curve_comparison.png
- precision_recall_curve_comparison.png
- model_comparison.csv